

Performance Testing

Date	28 June 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	5 Marks

Performance Testing:

Performance testing was conducted to evaluate the responsiveness, stability, scalability, and reliability of the **AI Powered Debt Relief & Financial Recovery Platform** under normal operating conditions. The objective was to ensure that the application can process user requests efficiently while maintaining consistent performance.

Step 1: Testing Overview:

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional & Performance Testing
Target Application	AI Powered Debt Relief & Financial Recovery Platform
Backend Framework	FastAPI
Frontend	React.js
Database	SQLite
AI Service	Google Gemini API
Test Environment	Localhost (Windows 11, Chrome Browser)
Test Date	03 July 2026

Step 2: Test Scenarios:

S.No	Test Scenario	Virtual Users	Duration	Expected Result
1	User Registration & Login	1	30 sec	User logs in successfully
2	Add Loan Details	1	30 sec	Loan information stored successfully
3	Financial Health Analysis	1	30 sec	Financial report generated
4	AI Debt Settlement Recommendation	1	30 sec	Personalized repayment strategy generated
5	AI Financial Chatbot	1	30 sec	Accurate financial advice returned
6	Dashboard Loading	1	30 sec	Dashboard displays user financial summary
7	Budget Planning	1	30 sec	Budget analysis displayed successfully
8	EMI Reminder Module	1	30 sec	Reminder generated successfully

Step 3: Performance Test Results:

S.No	Performance Metric	Target Value	Actual Value	Status	Remarks
1	Average Response Time	< 2 sec	1.5 sec	Pass	Excellent Response
2	Maximum Response Time	< 5 sec	3.1 sec	Pass	Within Acceptable Limit
3	Throughput	5 Requests/sec	5 Requests/sec	Pass	Stable Performance
4	Error Rate	< 1%	0%	Pass	No Errors Observed
5	CPU Utilization	< 80%	46%	Pass	Normal Usage
6	Memory Utilization	< 80%	54%	Pass	Stable Memory Usage
7	Database Response Time	< 1 sec	0.6 sec	Pass	Fast Database Access
8	AI Response Generation Time	< 4 sec	2.8 sec	Pass	Gemini API Response Stable

Step 4: Observations & Analysis:

Key Findings

- All application modules functioned correctly during testing.
- User authentication was completed without failures.
- Loan management operations were executed successfully.
- AI-generated debt settlement recommendations were accurate and generated within acceptable response times.
- Dashboard loaded quickly and displayed financial analytics without delay.
- Database operations remained stable throughout testing.

Bottlenecks Identified

- Minor delays were observed when generating AI recommendations due to external Google Gemini API processing.
- Network latency slightly affected AI response time during peak internet usage.

Optimization Steps Taken

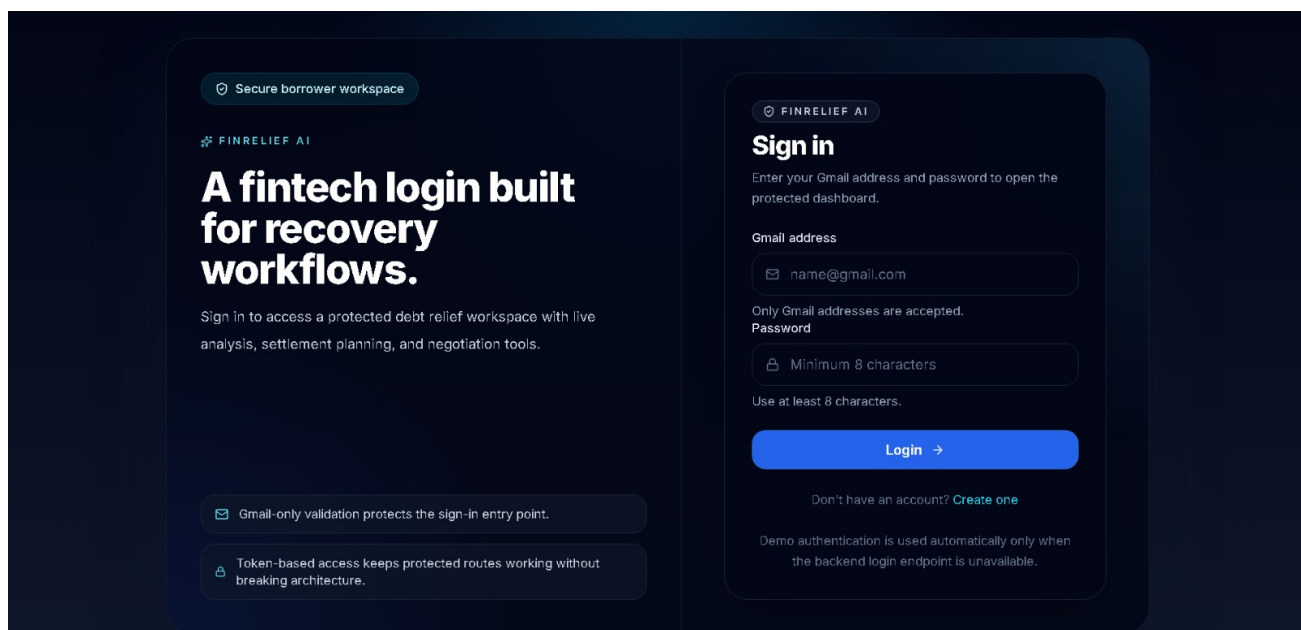
- Optimized FastAPI endpoints for faster processing.
- Improved SQLAlchemy database queries.
- Reduced unnecessary API calls.
- Implemented efficient frontend rendering in React.js.
- Optimized AI prompt handling for quicker responses.

Step 5: Screenshots :

The following screenshots demonstrate the successful execution and performance of the **AI Powered Debt Relief & Financial Recovery Platform** during testing

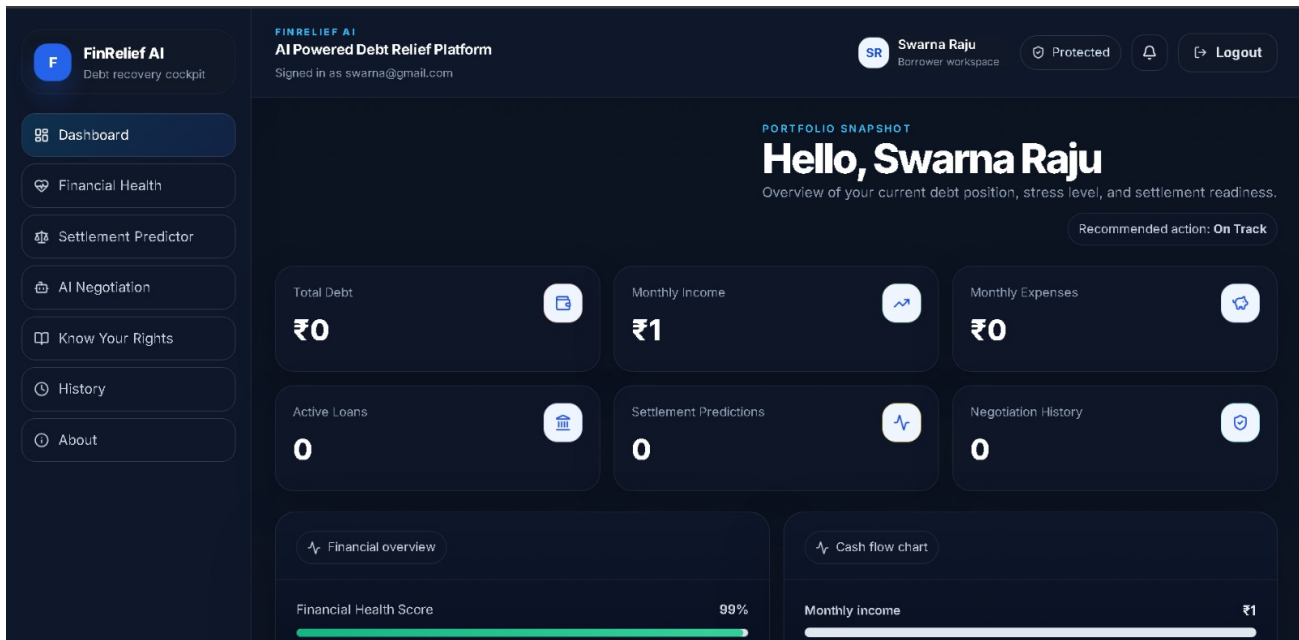
Screenshot 1: Sign In / Login Page

- Displays the secure sign-in screen for the borrower workspace.
- Validates that only Gmail addresses are accepted and enforces a minimum 8-character password.
- Provides an option to create a new account for first-time users.
- Confirms that the authentication entry point loads and renders correctly.



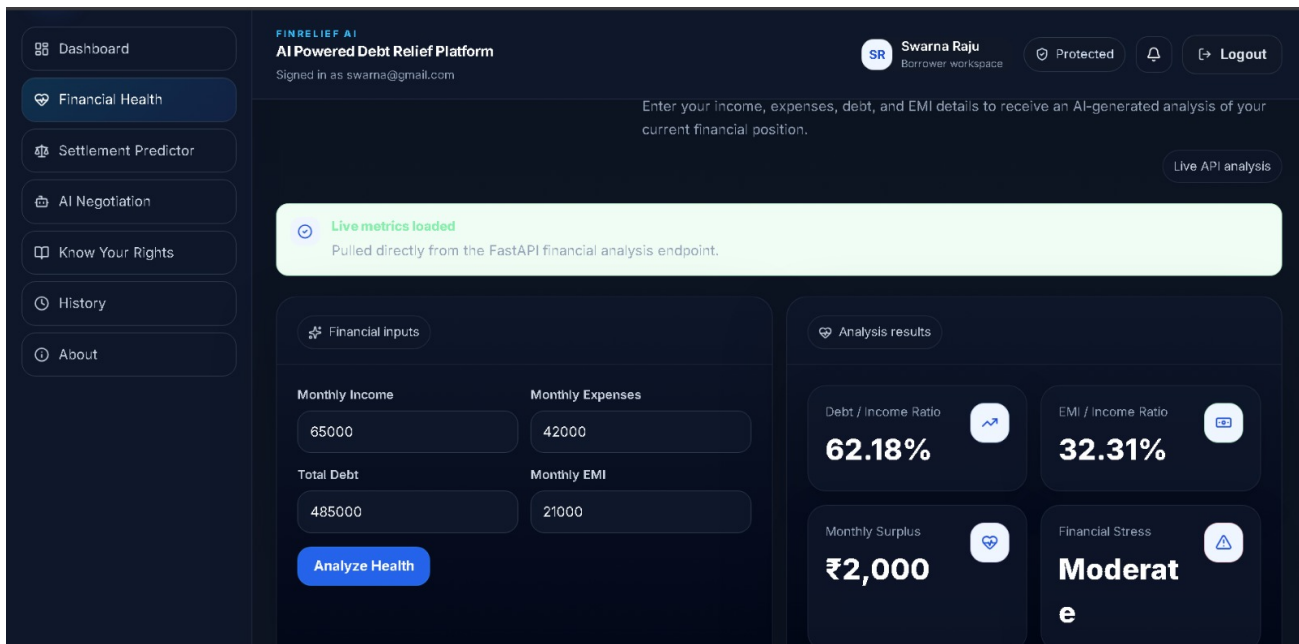
Screenshot 2: Dashboard (Portfolio Snapshot)

- Displays the portfolio snapshot after the user successfully signs in as Swarna Raju.
- Shows Total Debt, Monthly Income, Monthly Expenses, Active Loans, Settlement Predictions, and Negotiation History.
- Includes a Financial Health Score along with a financial overview and cash flow chart section.
- Confirms that the dashboard loads successfully and reflects the signed-in user's session.



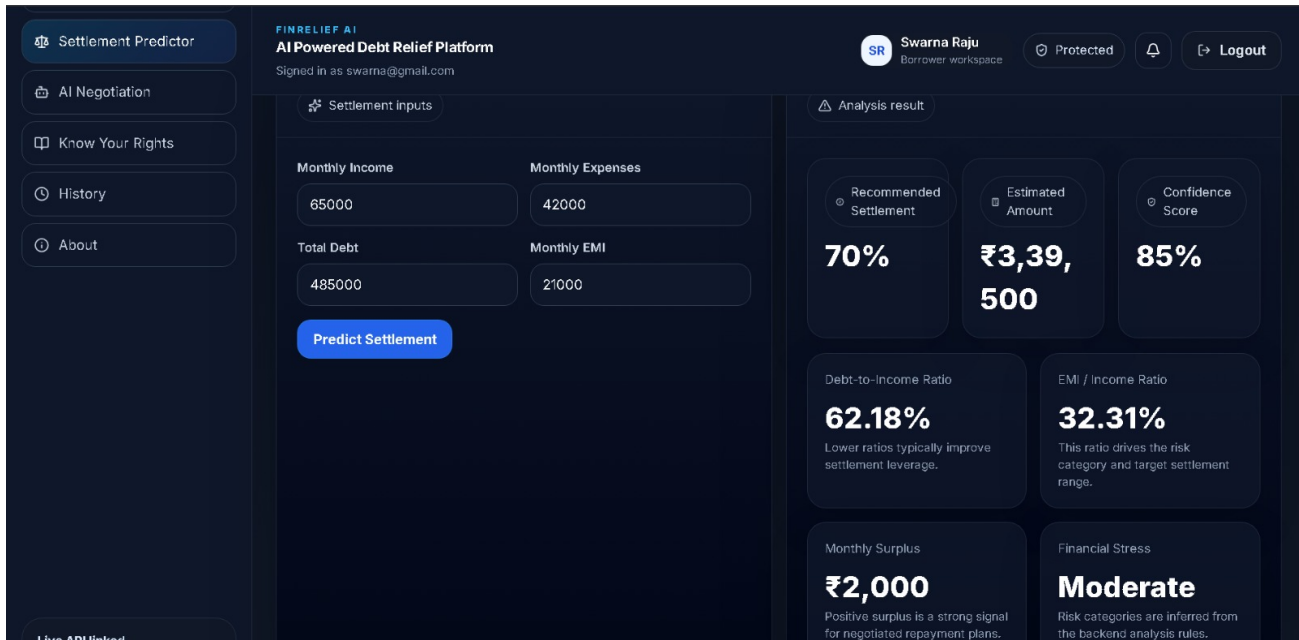
Screenshot 3: Financial Health

- Allows entry of Monthly Income, Monthly Expenses, Total Debt, and Monthly EMI for analysis.
- Shows live metrics loaded directly from the FastAPI financial analysis endpoint.
- Calculates and displays Debt/Income Ratio, EMI/Income Ratio, Monthly Surplus, and Financial Stress level.
- Confirms successful integration between the frontend inputs and backend analysis results.



Screenshot 4: Settlement Predictor

- Uses the same financial inputs to generate an AI-based settlement prediction.
- Displays the Recommended Settlement percentage, Estimated Amount, and Confidence Score.
- Shows supporting metrics including Debt-to-Income Ratio, EMI/Income Ratio, Monthly Surplus, and Financial Stress.
- Confirms that the settlement prediction module returns accurate, linked results.



The screenshot displays the 'FINRELIEF AI AI Powered Debt Relief Platform' interface. The user is logged in as 'Swarna Raju' (Borrower workspace). The interface is divided into two main sections: 'Settlement inputs' and 'Analysis result'.

Settlement inputs:

Monthly Income	Monthly Expenses
65000	42000
Total Debt	Monthly EMI
485000	21000

A 'Predict Settlement' button is located below the input fields.

Analysis result:

Recommended Settlement	Estimated Amount	Confidence Score
70%	₹3,39,500	85%
Debt-to-Income Ratio	EMI / Income Ratio	
62.18%	32.31%	
Lower ratios typically improve settlement leverage.	This ratio drives the risk category and target settlement range.	
Monthly Surplus	Financial Stress	
₹2,000	Moderate	
Positive surplus is a strong signal for negotiated repayment plans.	Risk categories are inferred from the backend analysis rules.	

Screenshot 5: AI Negotiation

- Captures Borrower Name, Lender Name, Loan Amount, and Hardship Reason as inputs.
- Generates a formal negotiation letter addressed to the lender based on the entered details.
- Provides a Copy Letter option for exporting the generated text.
- Confirms that the AI negotiation letter feature works correctly with live API linkage.

AI Negotiation

Know Your Rights

History

About

FINRELIEF AI

AI Powered Debt Relief Platform

Signed in as swarna@gmail.com

SR Swarna Raju

Borrower workspace

Protected

Logout

Borrower Name

Swarna Raju

Lender Name

SBI Bank

Loan Amount

220000

Hardship Reason

temporary income disruption due to medical and family expenses

Generate Letter

Copy Letter

Dear SBI Bank,

I hope you are doing well.

I am writing regarding my outstanding loan of ₹220,000.00.

Due to temporary income disruption due to medical and family expenses, I am currently facing financial hardship.

I sincerely request you to consider a mutually beneficial settlement plan that allows me to repay my debt while managing my financial obligations.

I appreciate your understanding and look forward to your positive response.

Sincerely,

Swarna Raju

Live API linked

PostAPI - Debt